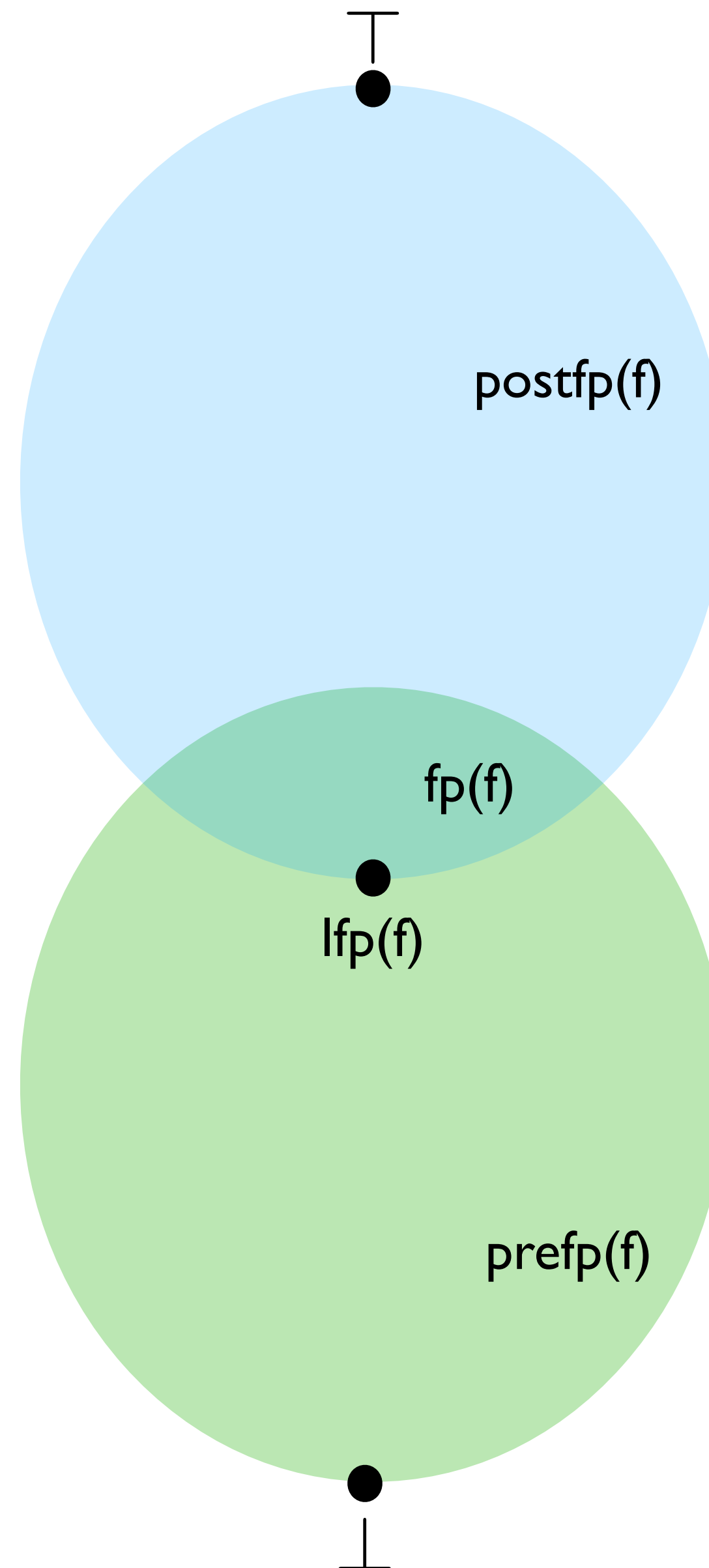


Static analysis and software verification

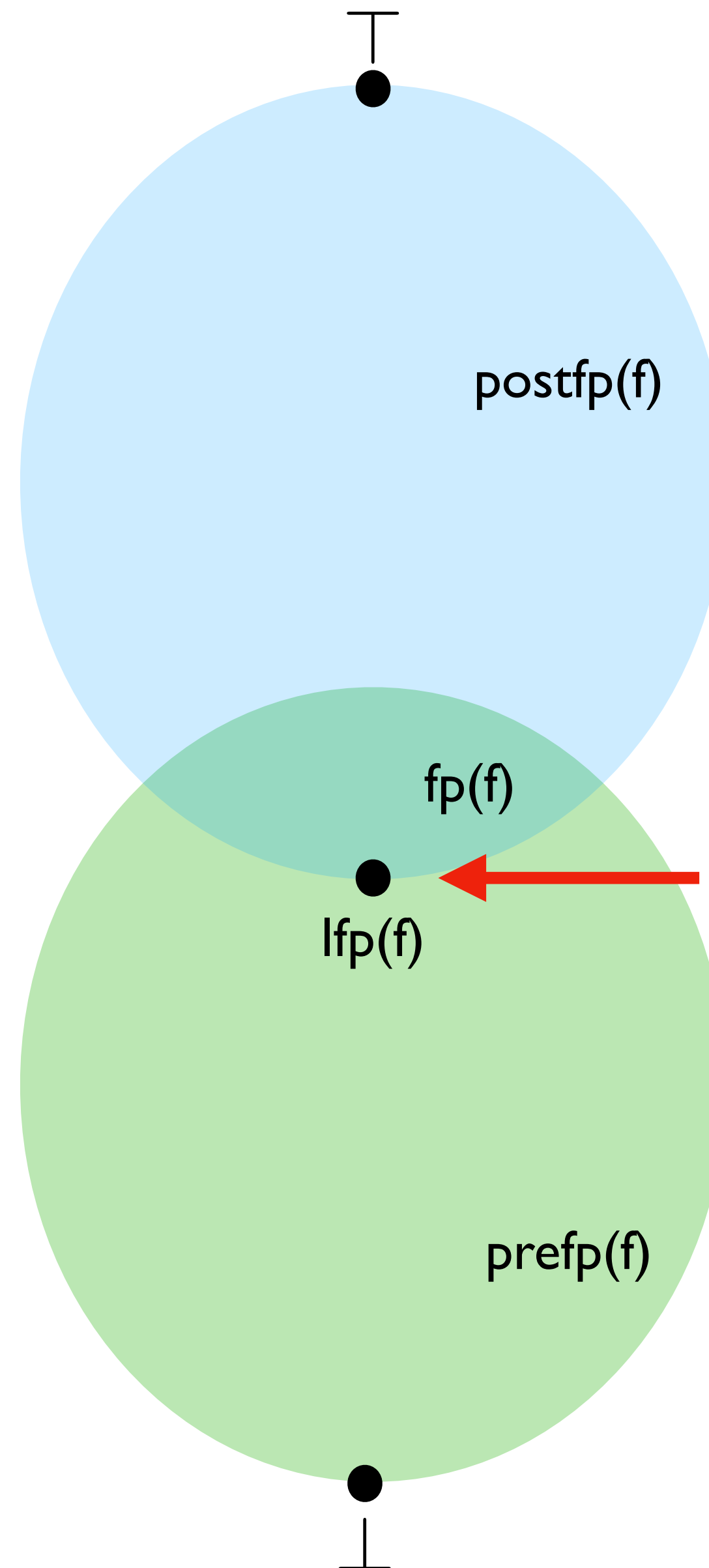
Lecture II - The interval abstract domain

The interval abstract domain

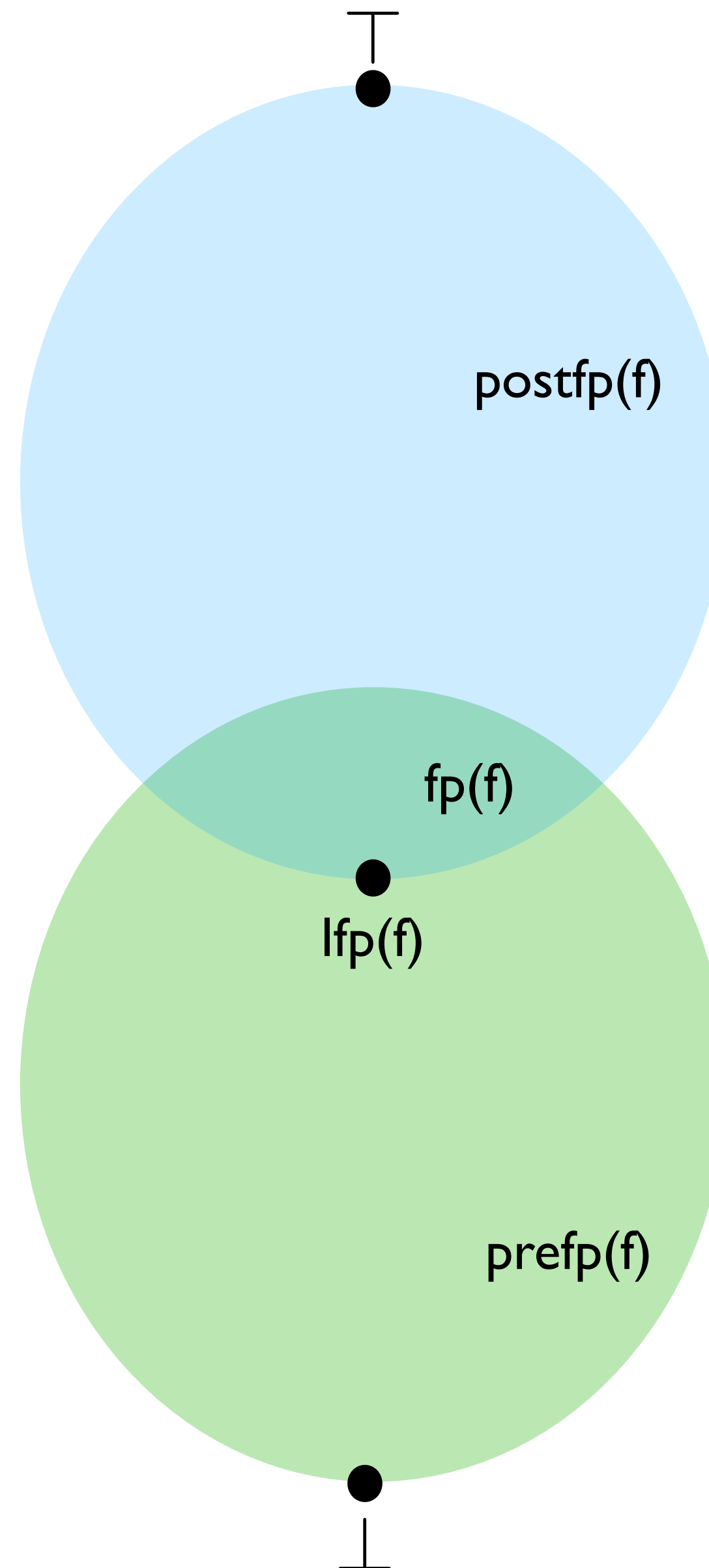
Widening



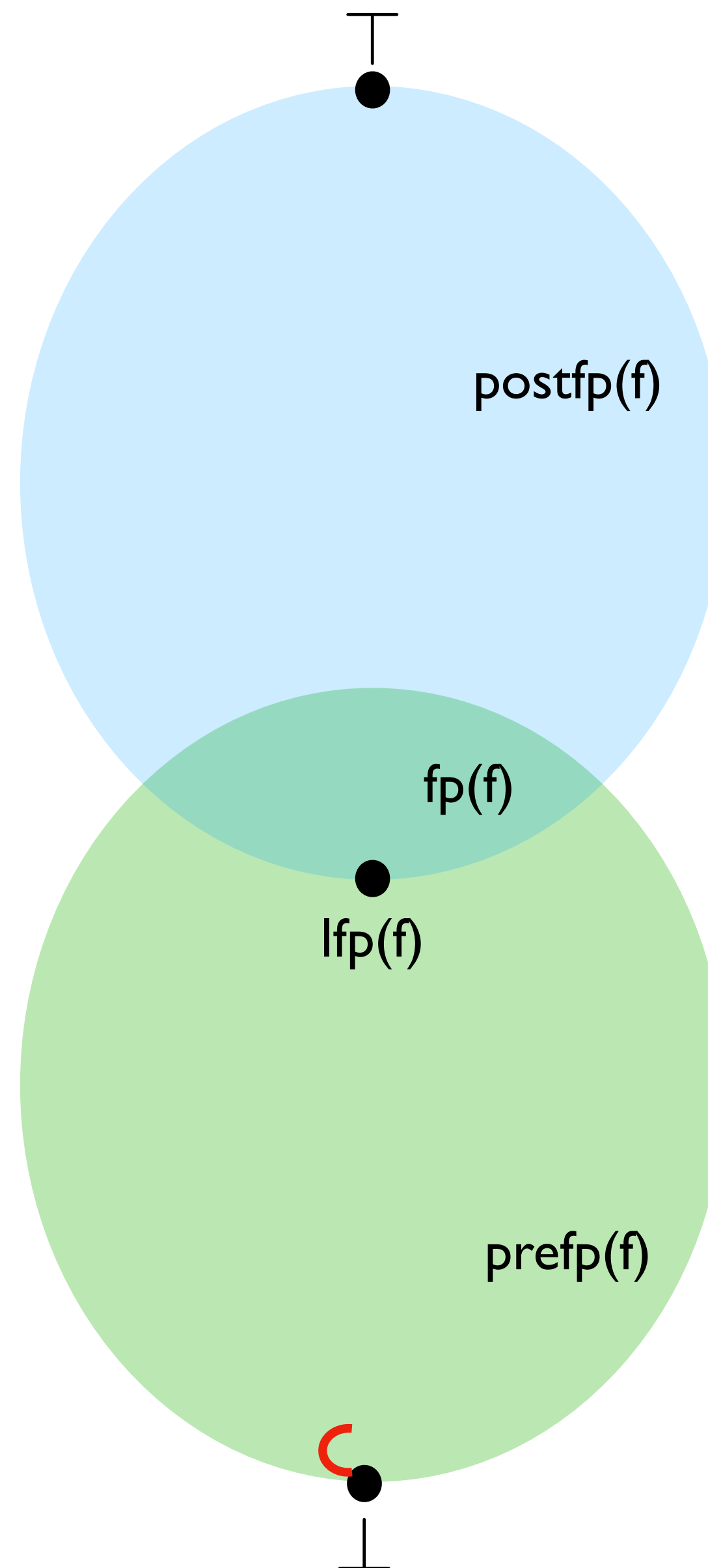
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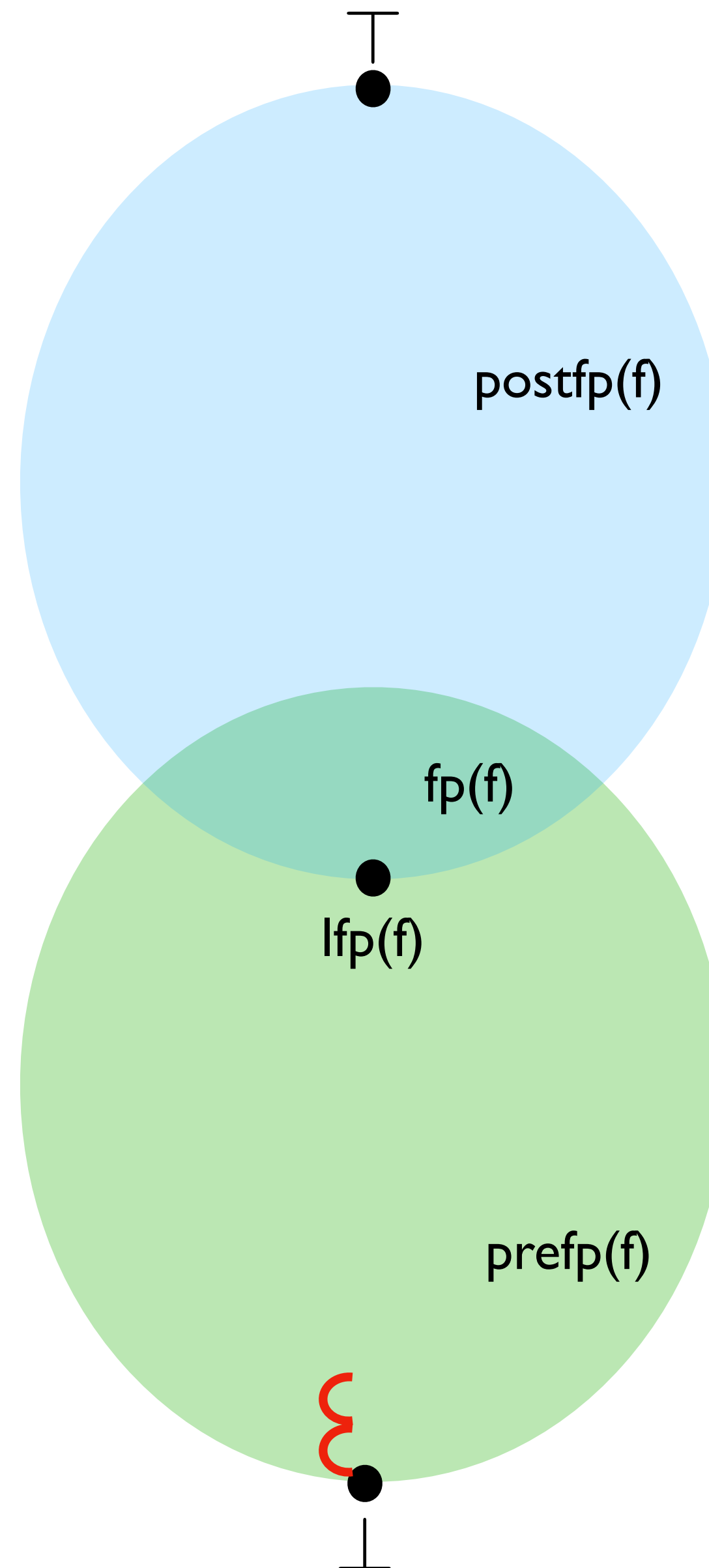
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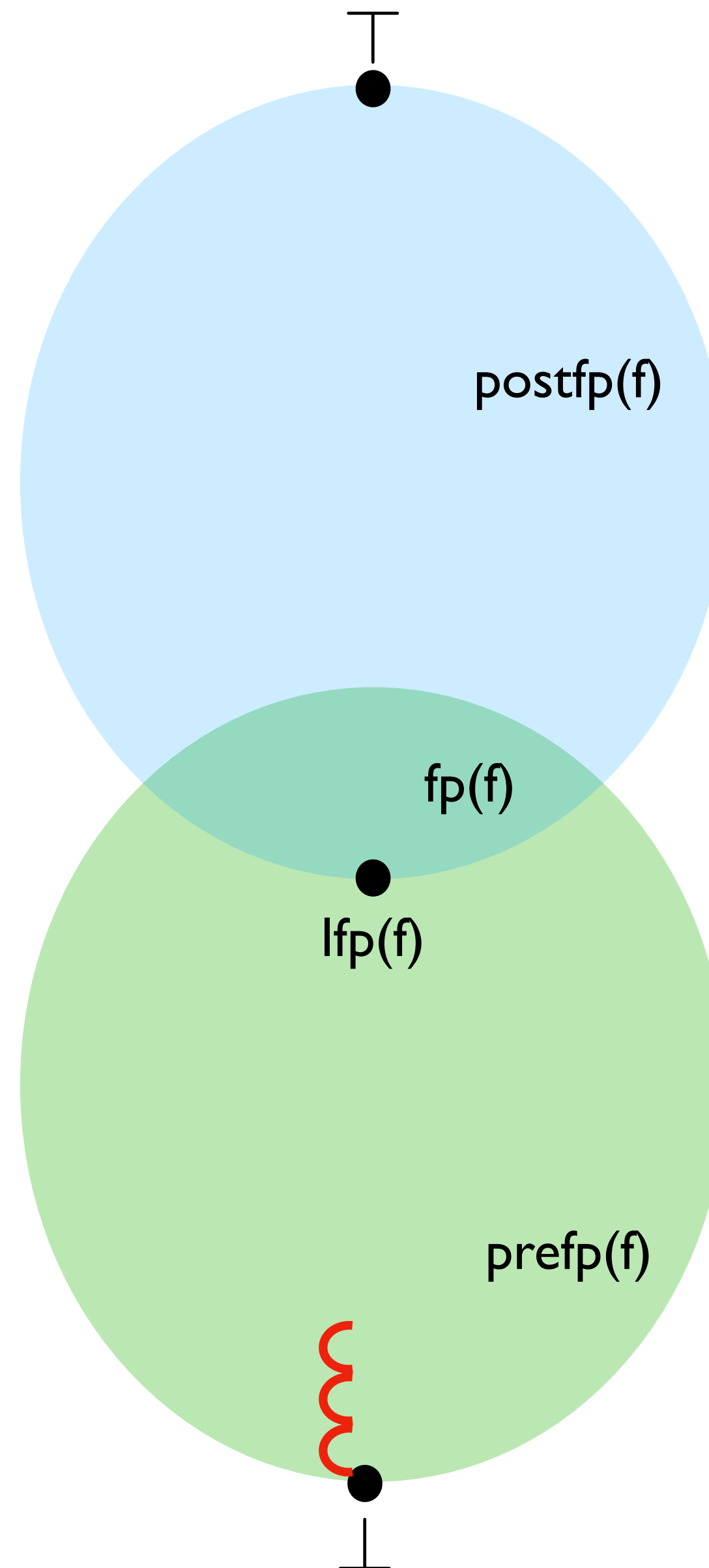
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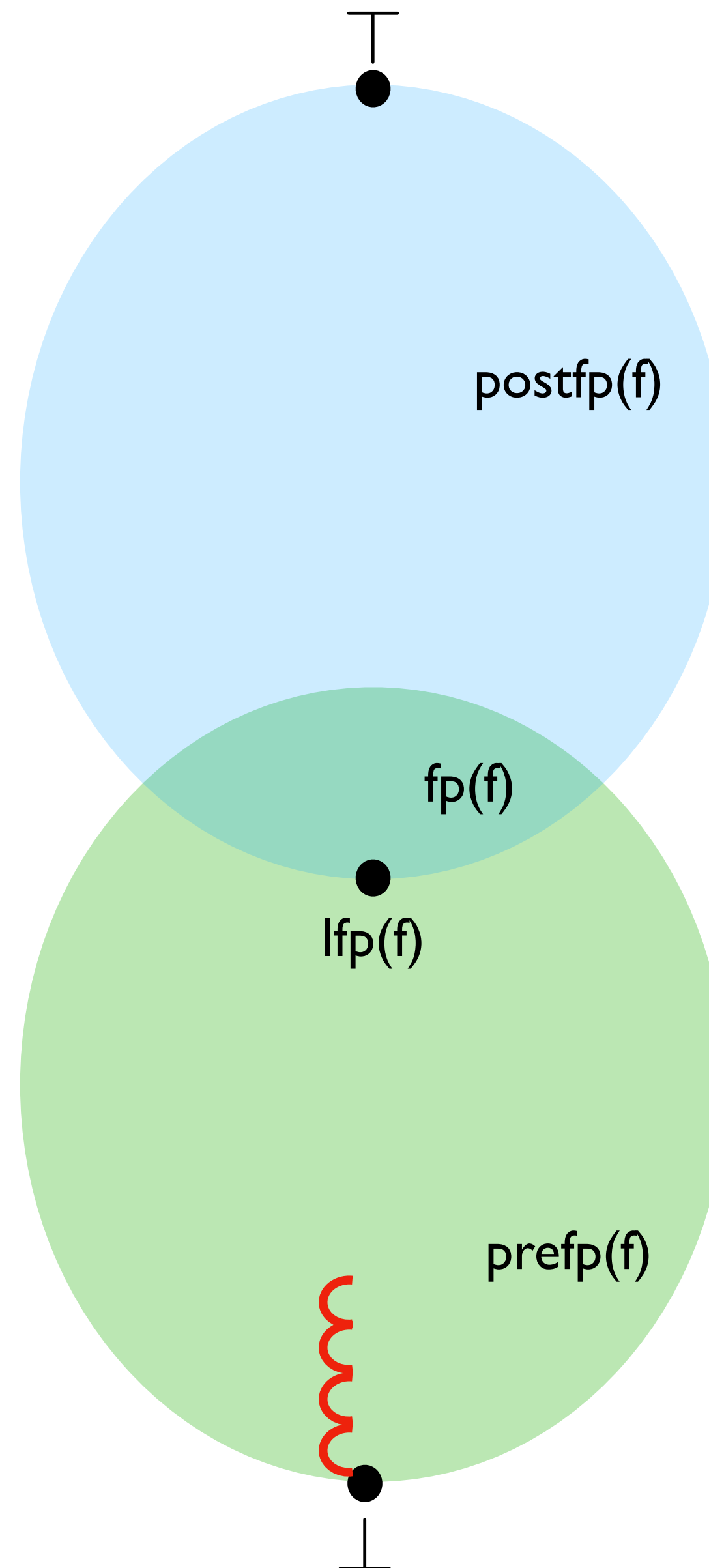
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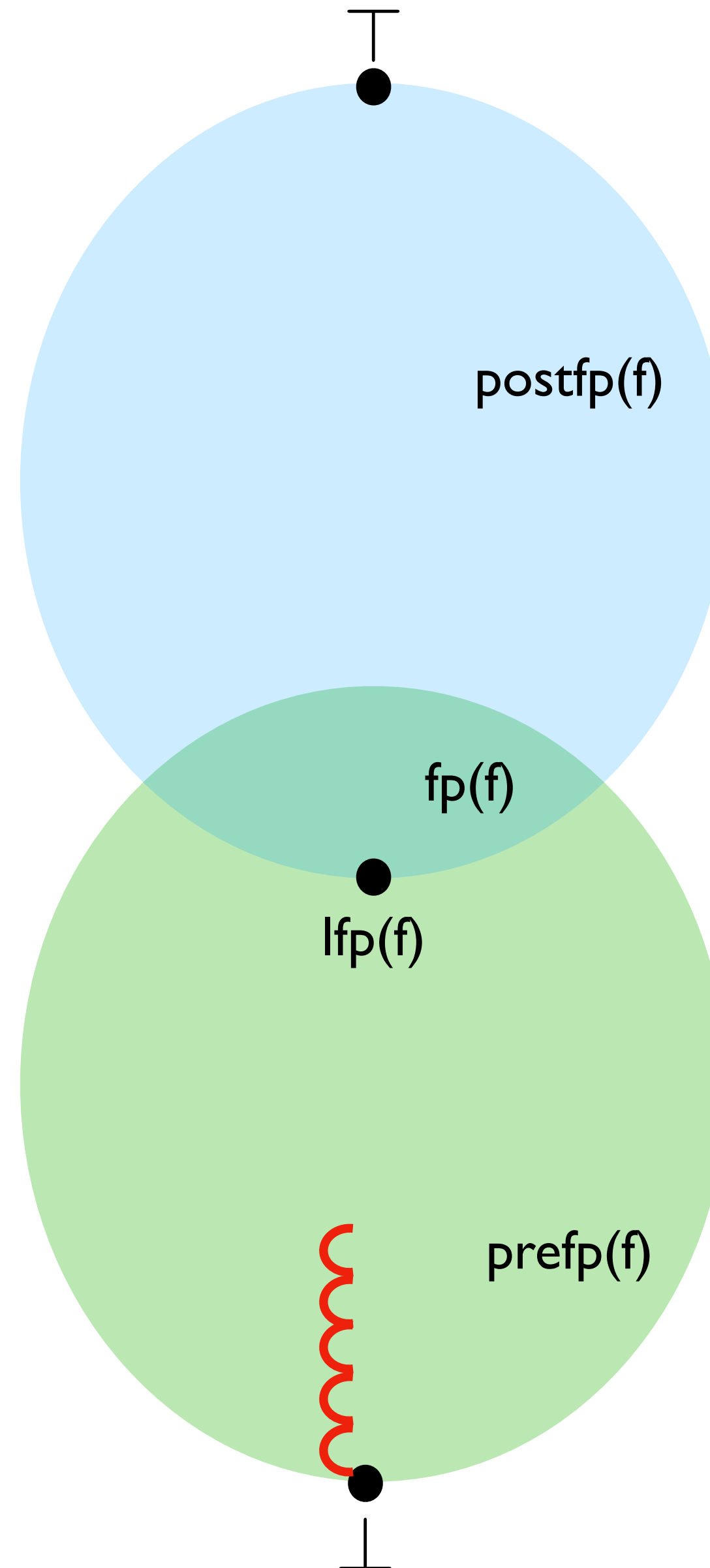
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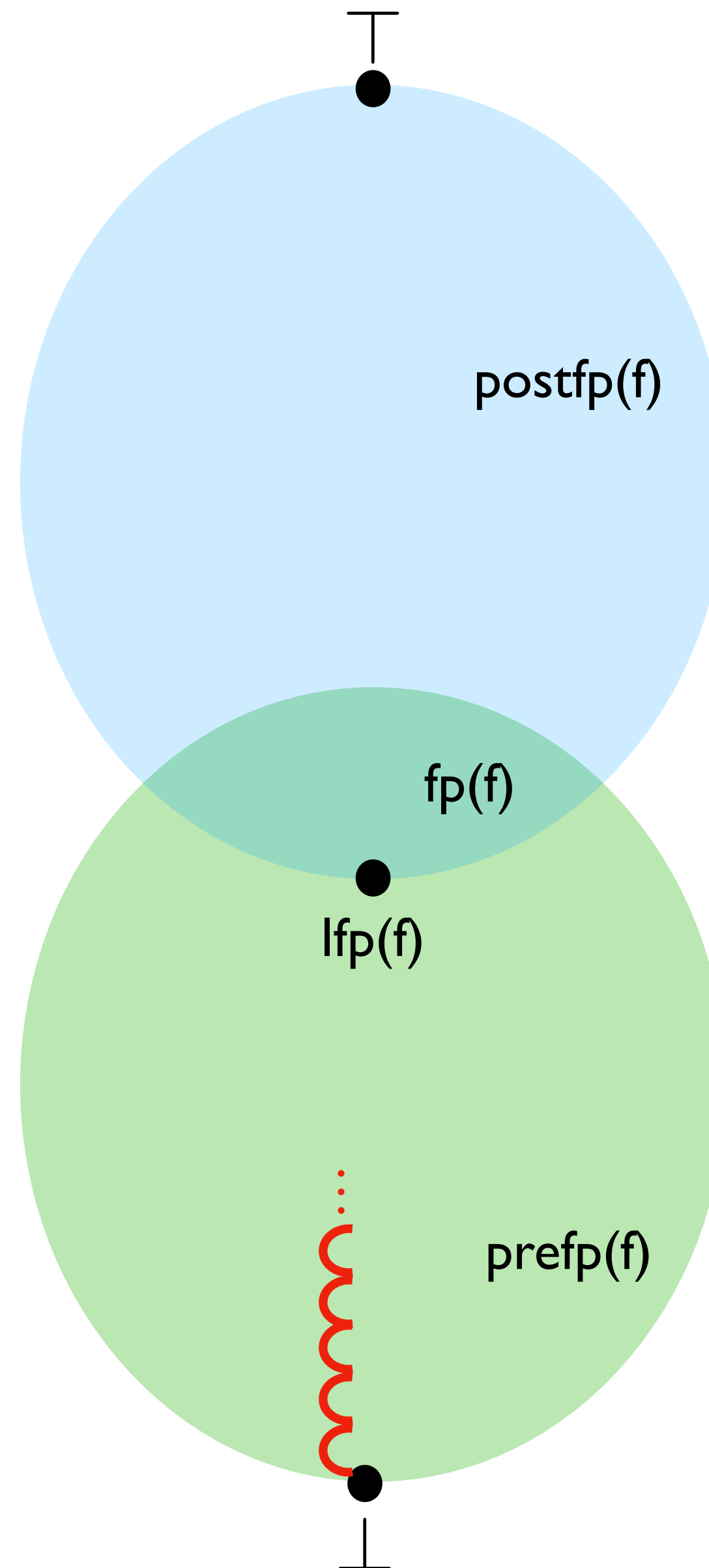
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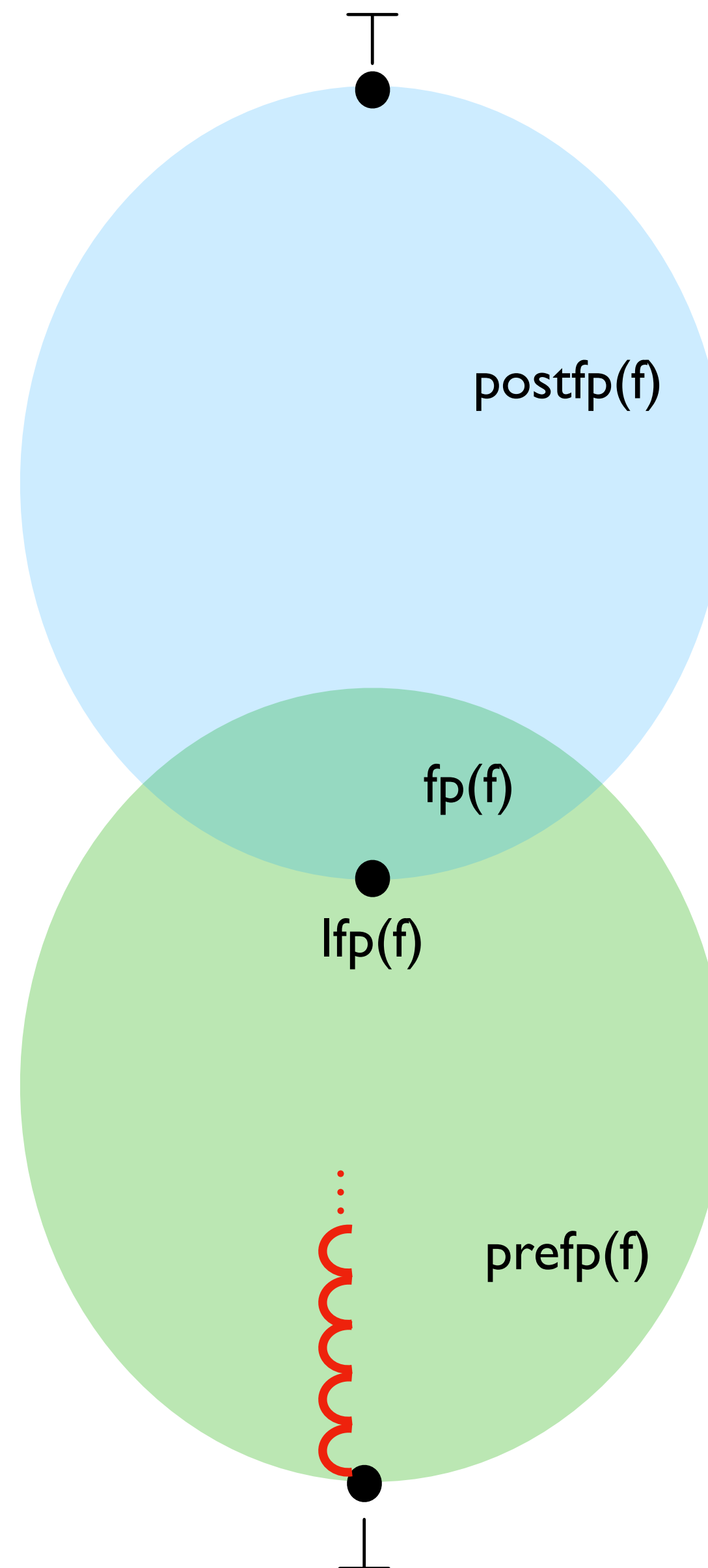
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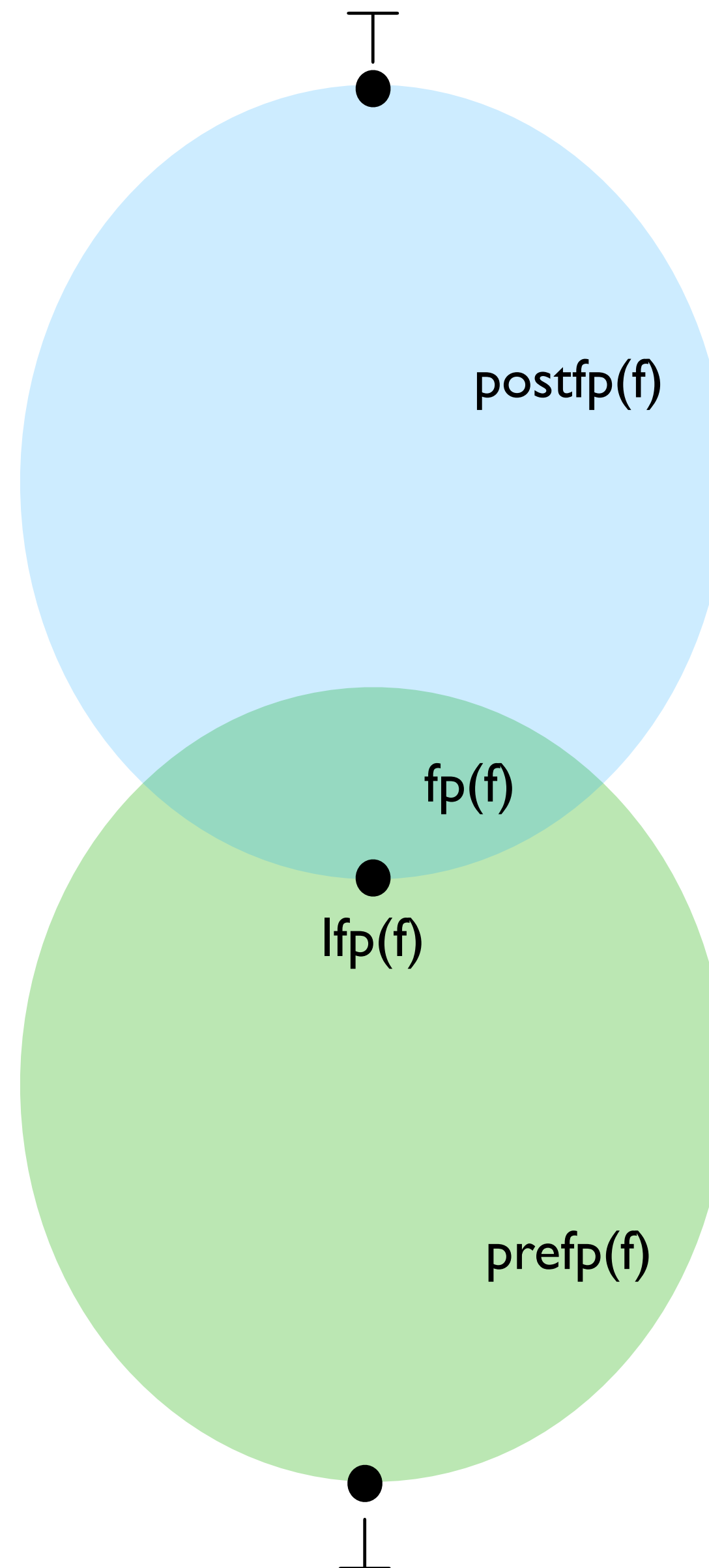


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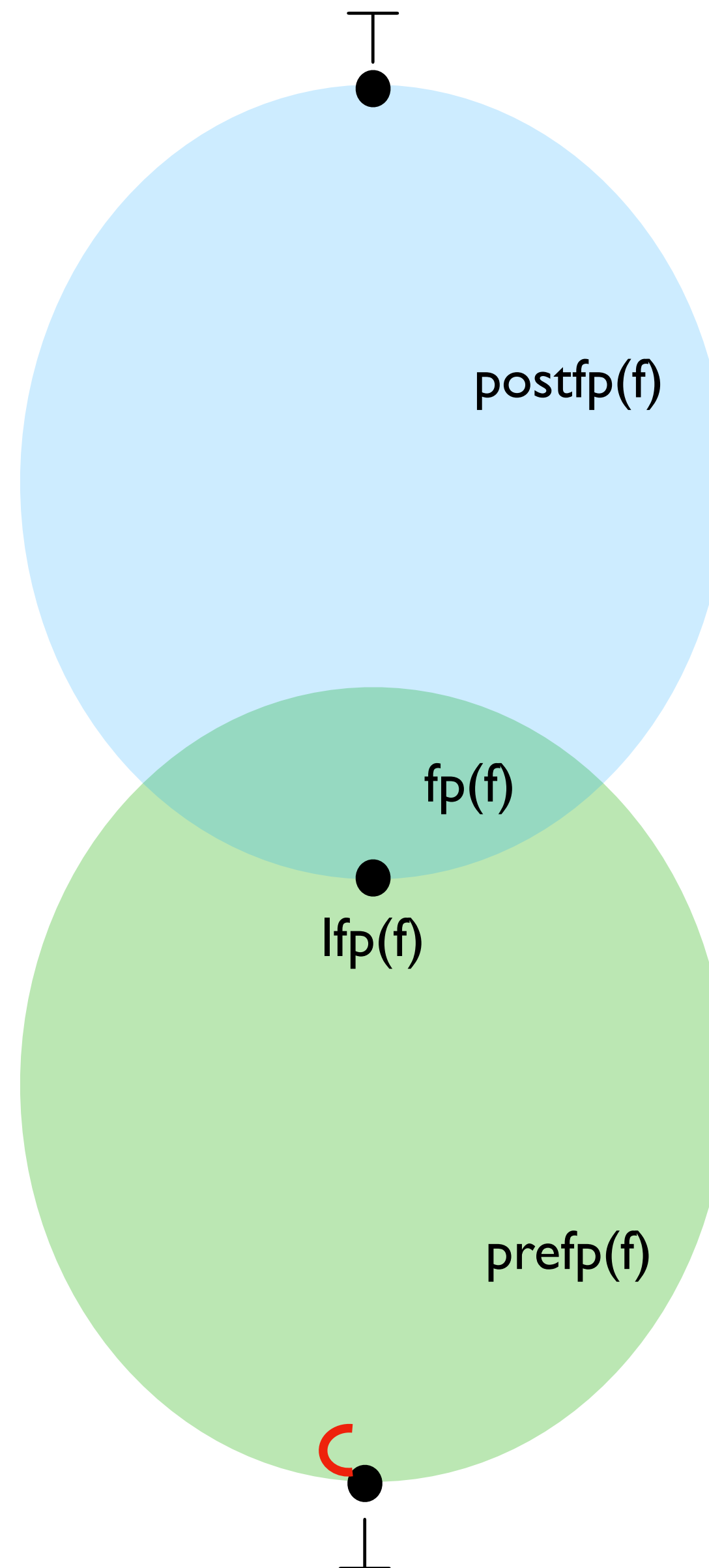


Divergence!

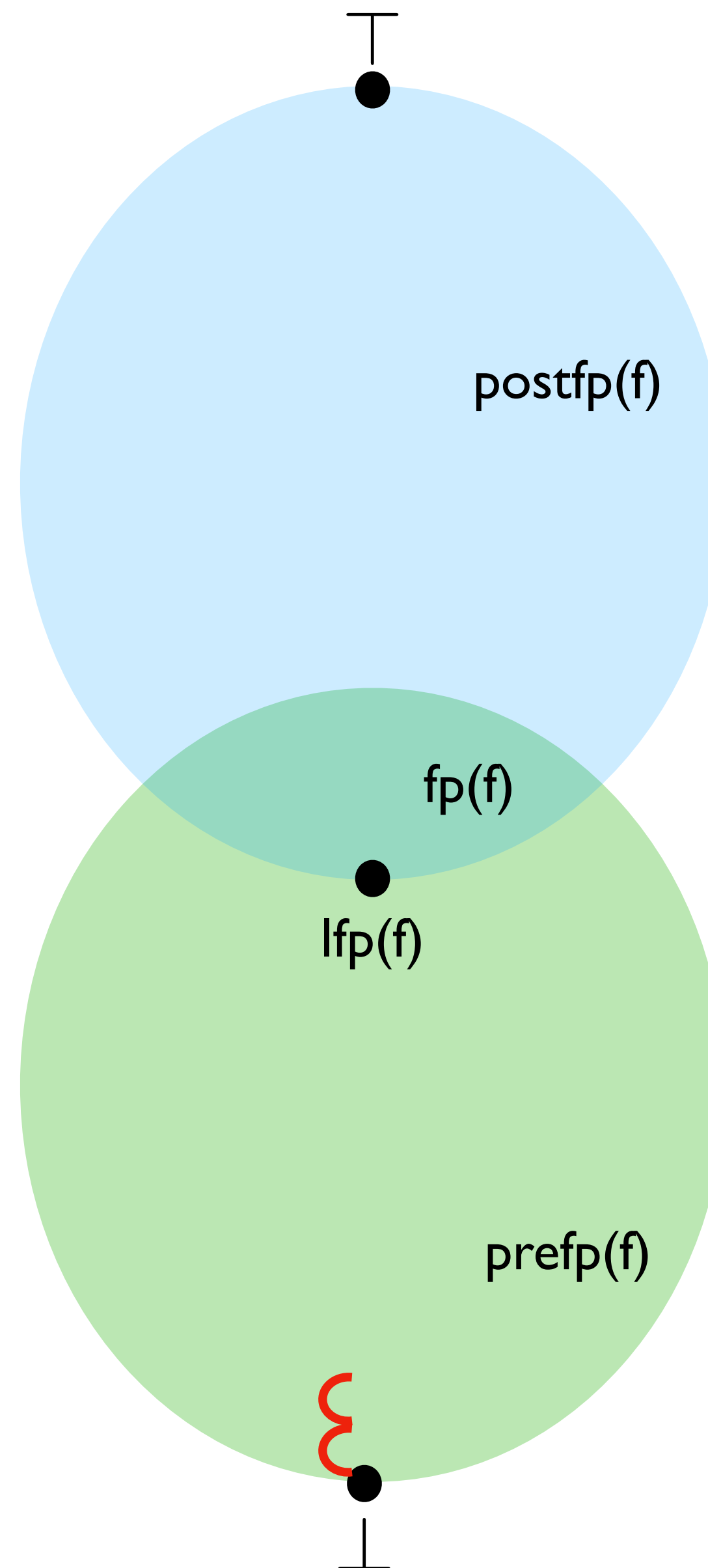
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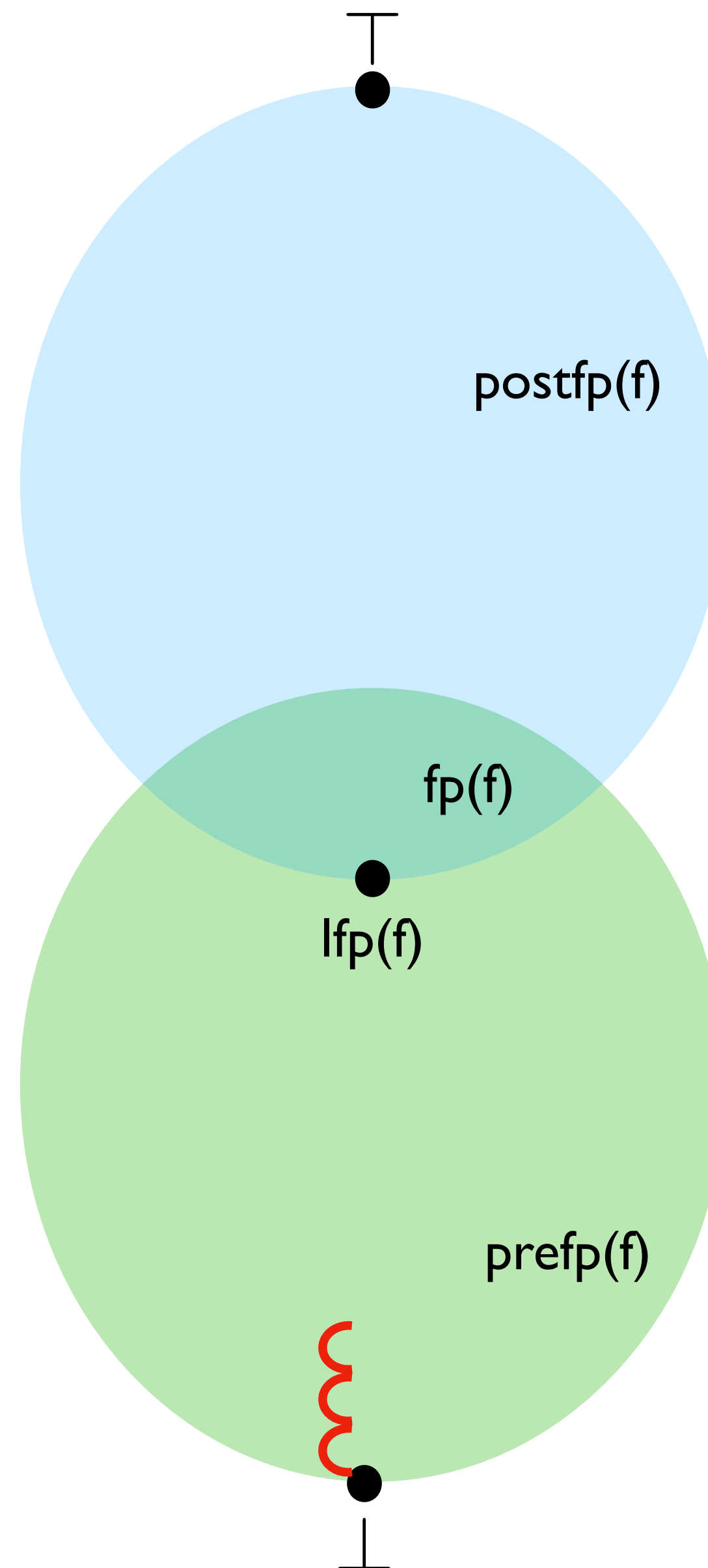
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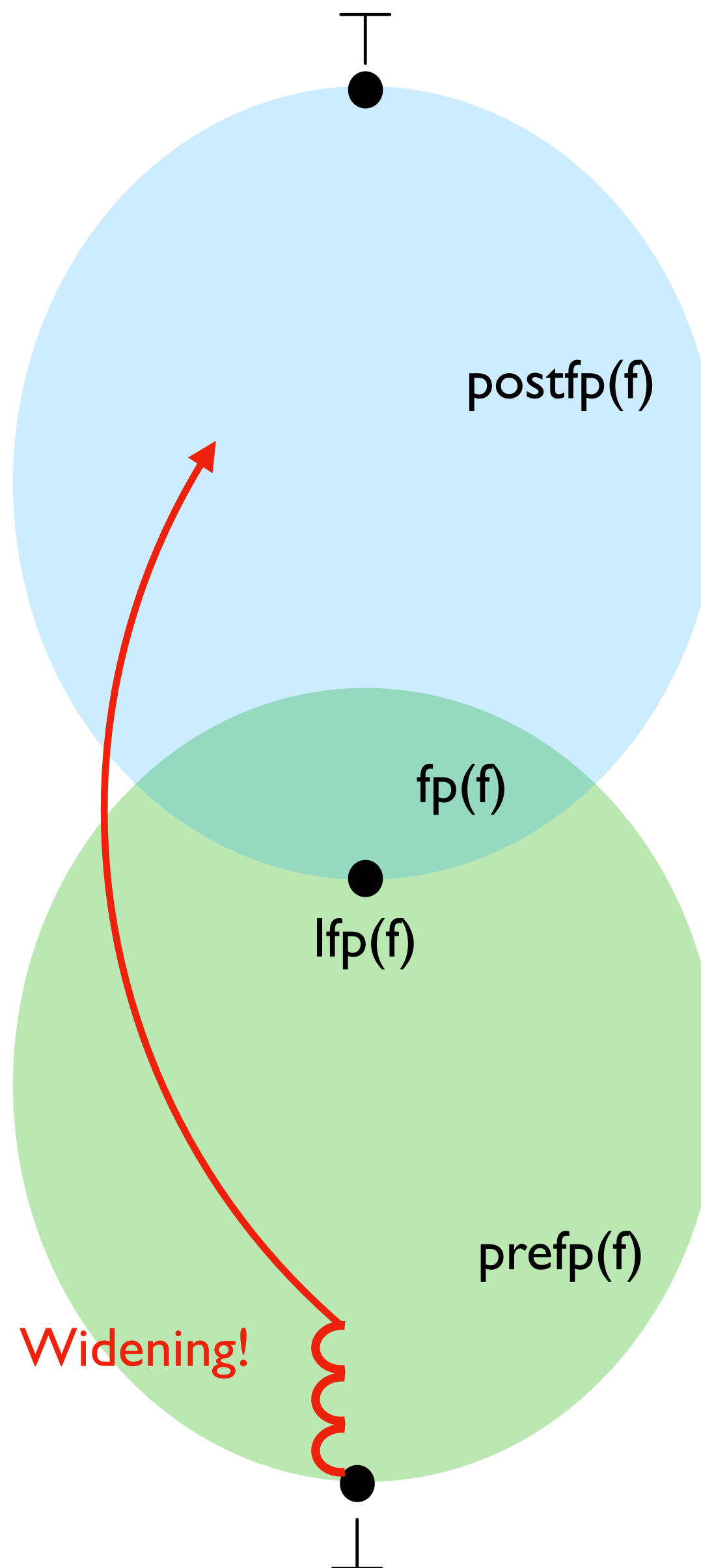
Widening



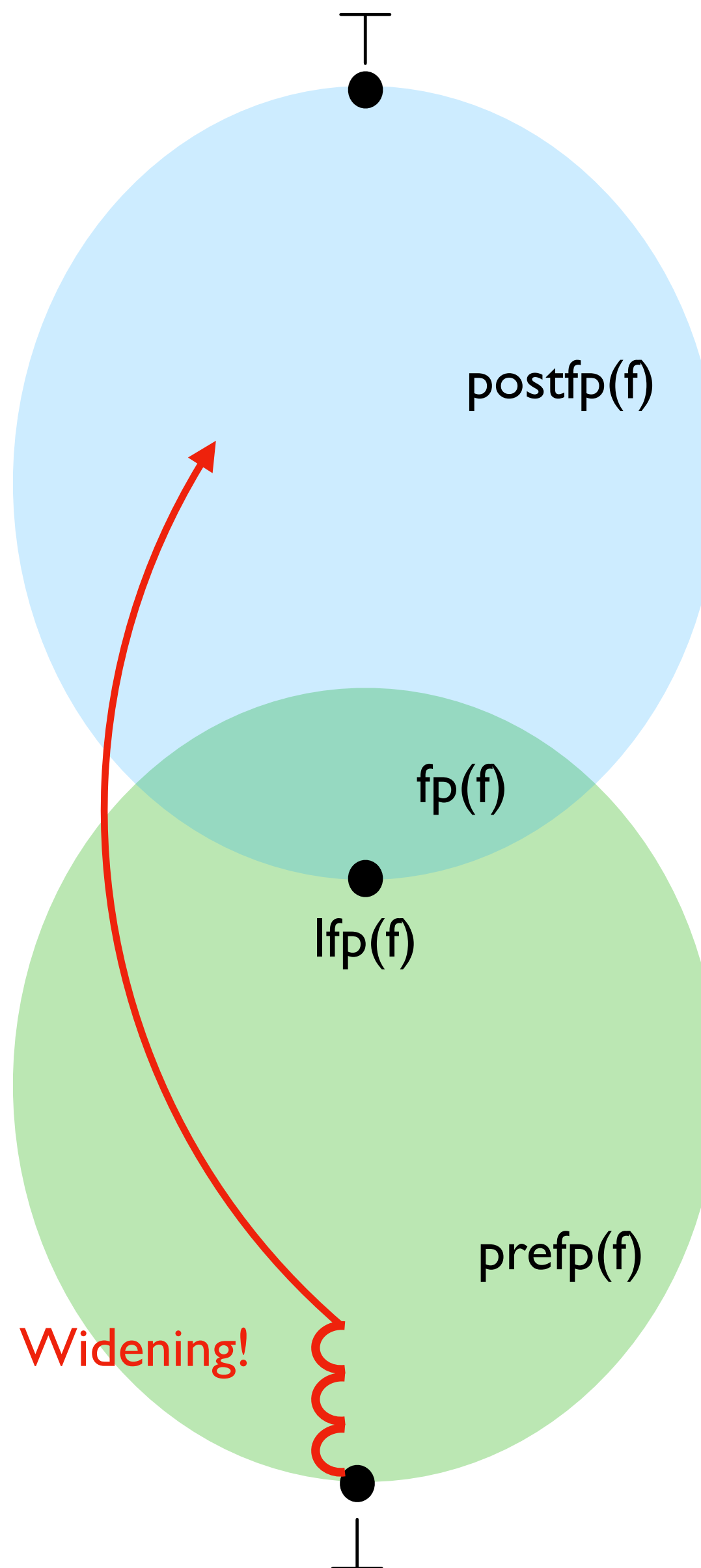
Widening



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Widening



Widening: over-approximates
the $\text{lfp}(f)$
but guarantees convergence

Widening

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Example of widening?

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$$x \nabla y = \begin{cases} x & \text{if } y \sqsubseteq x \\ \top & \text{otherwise} \end{cases}$$

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Naive

Widening for intervals

$$[a, b] \nabla [c, d] = \left[\begin{array}{ll} \begin{cases} a & \text{if } a \leq c \\ -\infty & \text{otherwise} \end{cases}, & \begin{cases} b & \text{if } b \geq d \\ +\infty & \text{otherwise} \end{cases} \right]$$

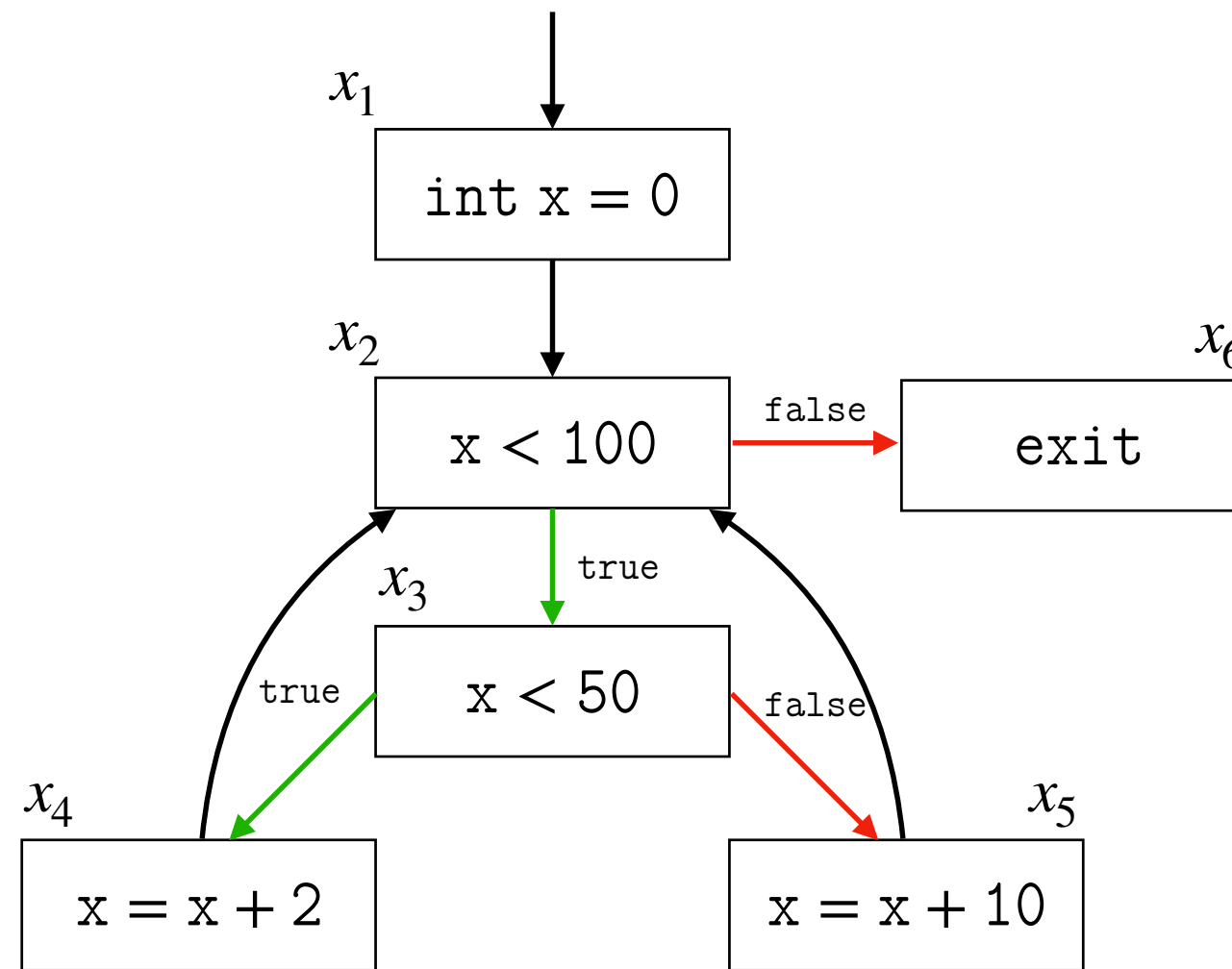
Widening for intervals

$$[a, b] \nabla [c, d] = \left[\begin{array}{ll} a & \text{if } a \leq c \\ -\infty & \text{otherwise} \end{array}, \begin{array}{ll} b & \text{if } b \geq d \\ +\infty & \text{otherwise} \end{array} \right]$$

- Not commutative: $[2,3] \nabla [1,4] = [-\infty, +\infty]$, $[1,4] \nabla [2,3] = [1,4]$
- Not monotone: let $[0,1] \sqsubseteq [0,3]$, $[0,1] \nabla [0,2] = [0, +\infty]$, $[0,3] \nabla [0,2] = [0,3]$

Widening

Where to apply it?



$$x_1 = \alpha(I)$$

$$x_2 = [[x = 0]]^\# x_1 \sqcup [[x = x + 2]]^\# x_4 \sqcup [[x = x + 10]]^\# x_5$$

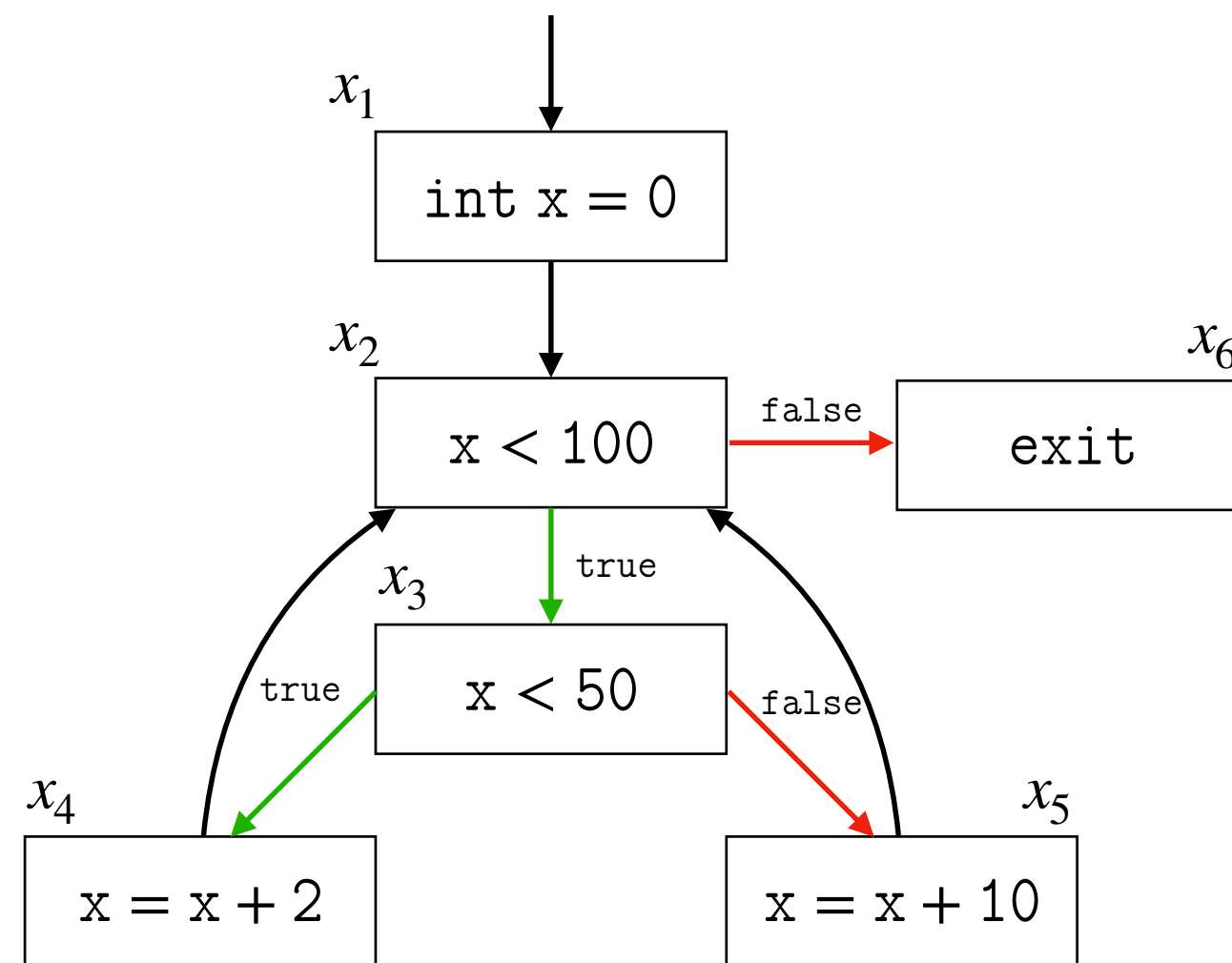
$$x_3 = [[x < 100]]^\# x_2$$

$$x_4 = [[x < 50]]^\# x_3$$

$$x_5 = [[x \geq 50]]^\# x_3$$

Widening

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Without widening

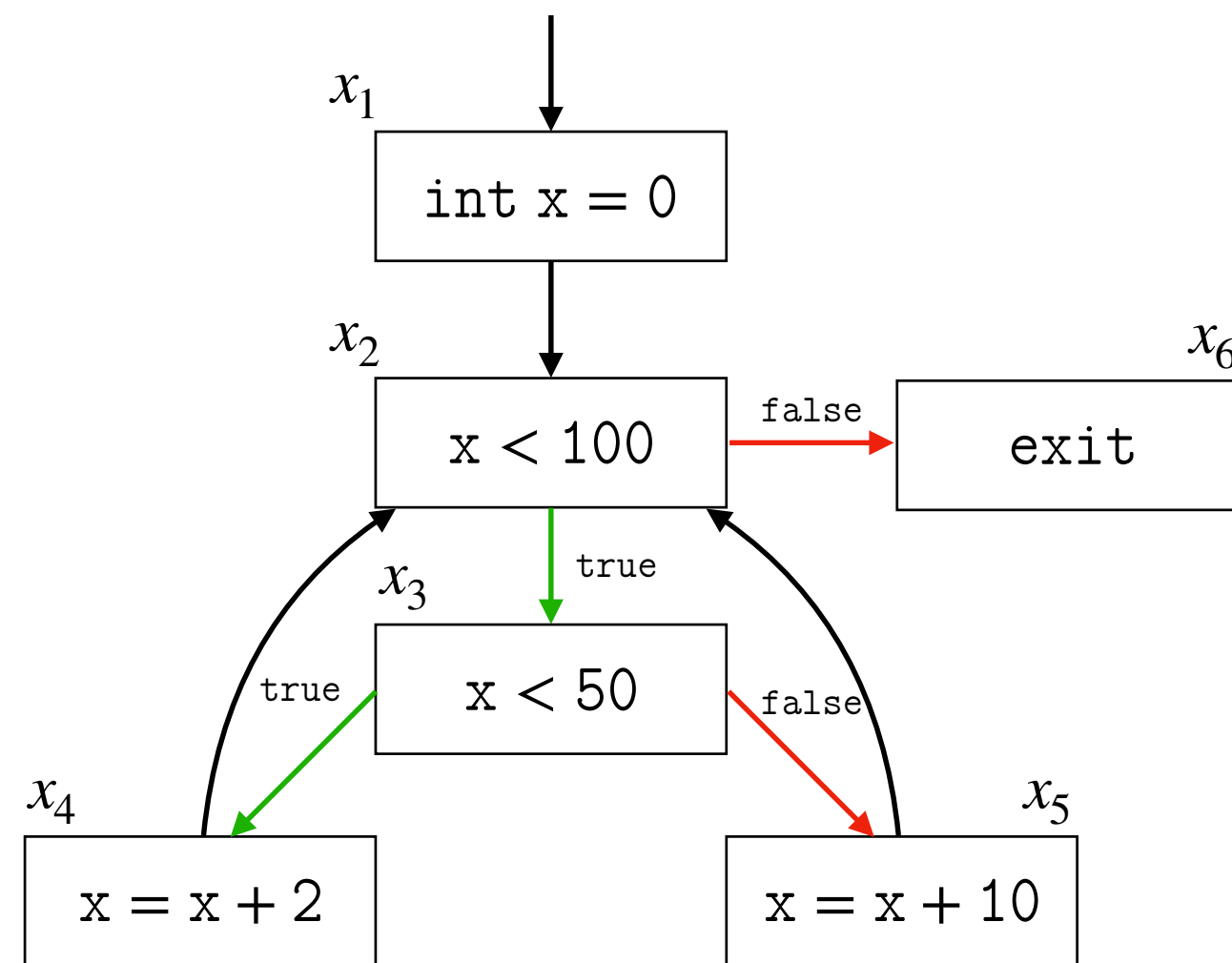
$$F^\# = \{x_i = x_i \sqcup f_i^\#(x_1, \dots, x_m) \mid i = 1, 2, \dots, m\}$$

With widening

$$F^\# = \{x_i = x_i \nabla f_i^\#(x_1, \dots, x_m) \mid i = 1, 2, \dots, m\}$$

Widening

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Without widening

$$F^\# = \{x_i = x_i \sqcup f_i^\#(x_1, \dots, x_m) \mid i = 1, 2, \dots, m\}$$

With widening (too coarse)

$$F^\# = \{x_i = x_i \nabla f_i^\#(x_1, \dots, x_m) \mid i = 1, 2, \dots, m\}$$

Widening

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- It is sufficient to apply the widening on one node in each cycle of the CFG (usually the entry node)

Widening

Where to apply it?

$$F^\# = \begin{cases} x_i = x_i \nabla f_i^\#(x_1, \dots, x_m) & \text{if } i \in WP \\ x_i = x_i \sqcup f_i^\#(x_1, \dots, x_m) & \text{otherwise} \end{cases}$$

$$x_1 = \alpha(I)$$

$$x_2 = [[x = 0]]^\# x_1 \sqcup [[x = x + 2]]^\# x_4 \sqcup [[x = x + 10]]^\# x_5$$

$$x_3 = [[x < 100]]^\# x_2$$

$$x_4 = [[x < 50]]^\# x_3$$

$$x_5 = [[x > = 50]]^\# x_3$$

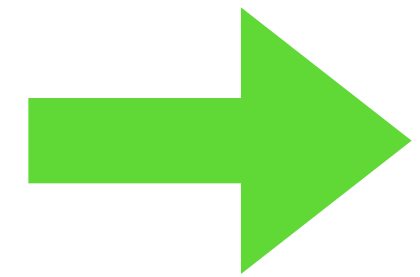
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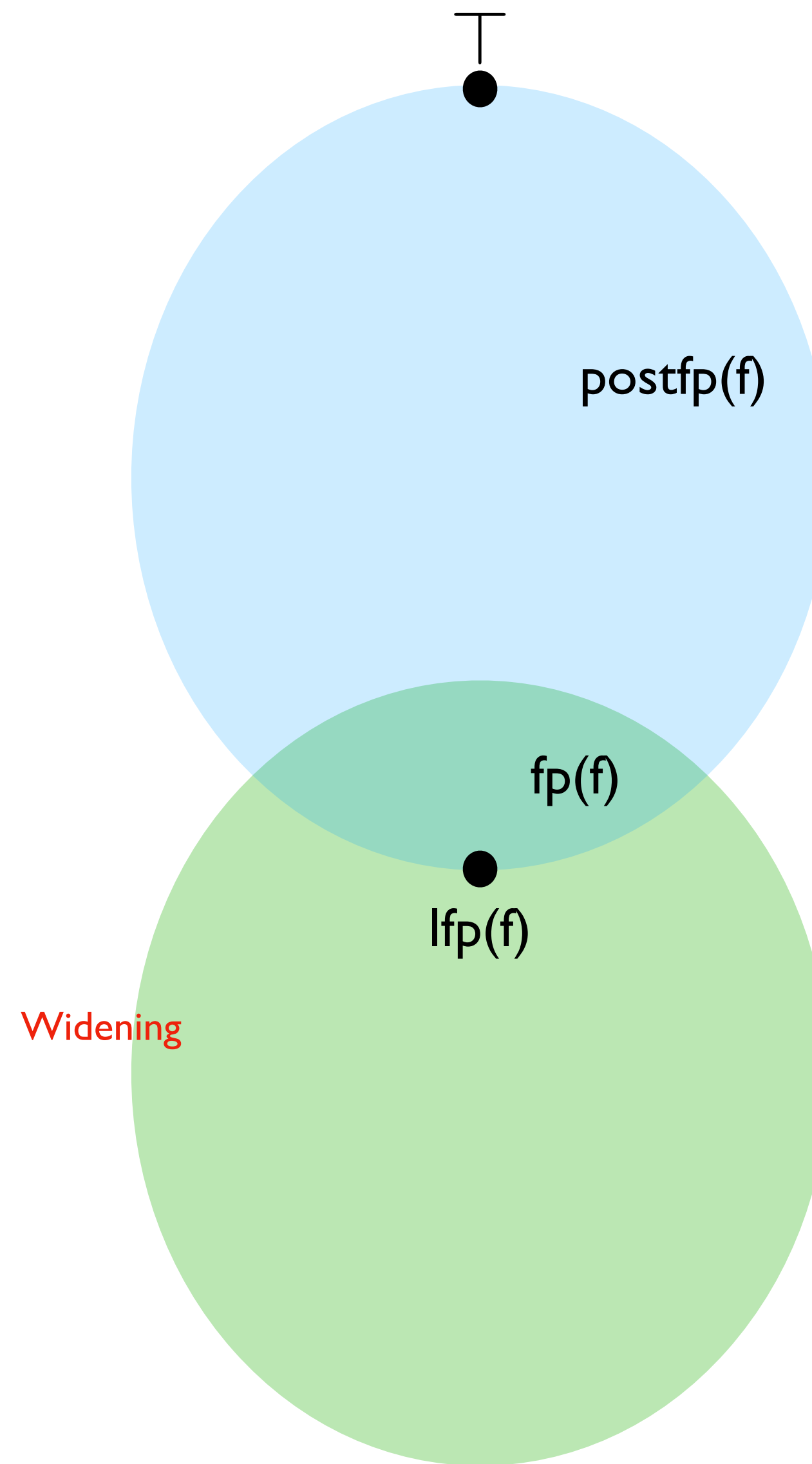


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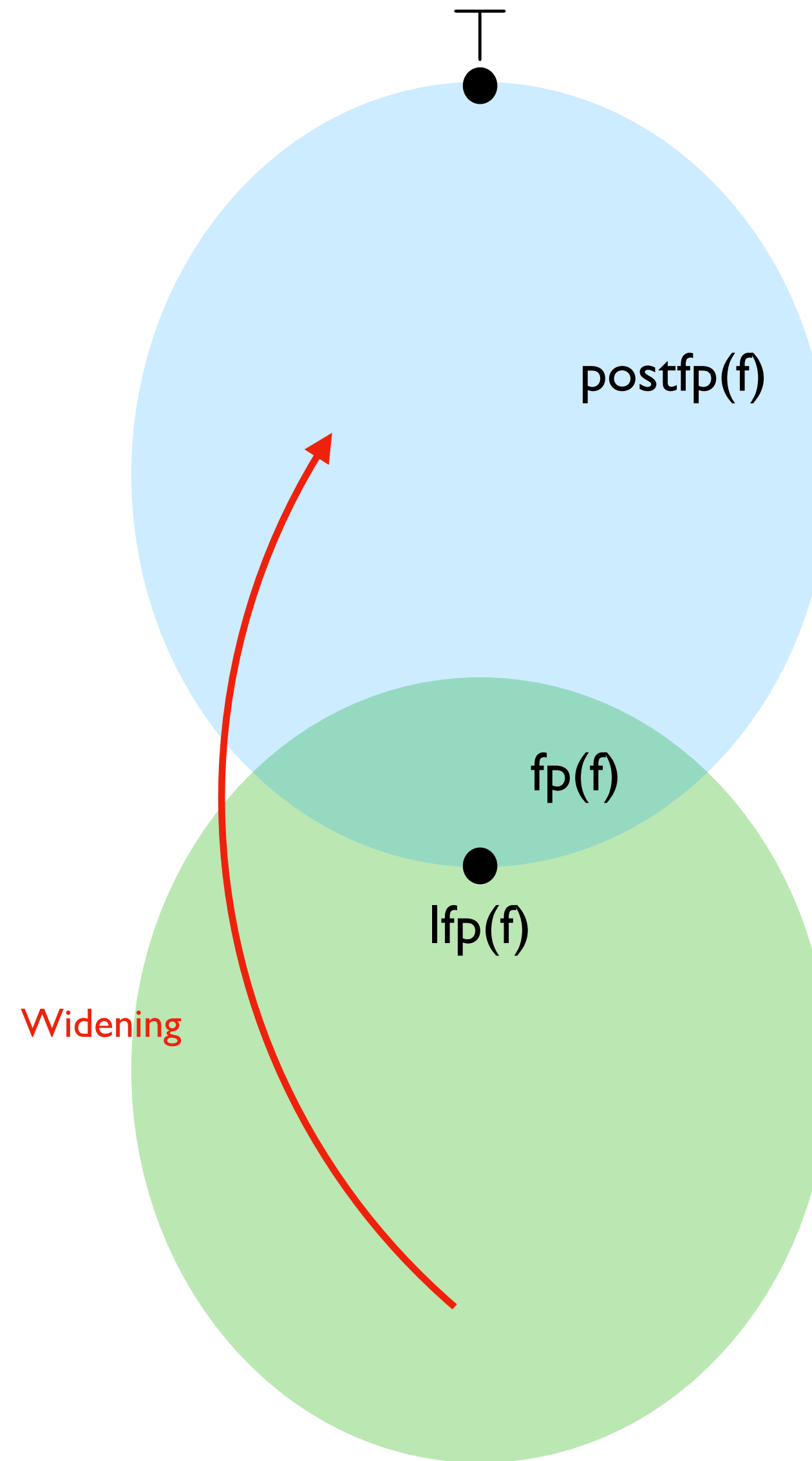
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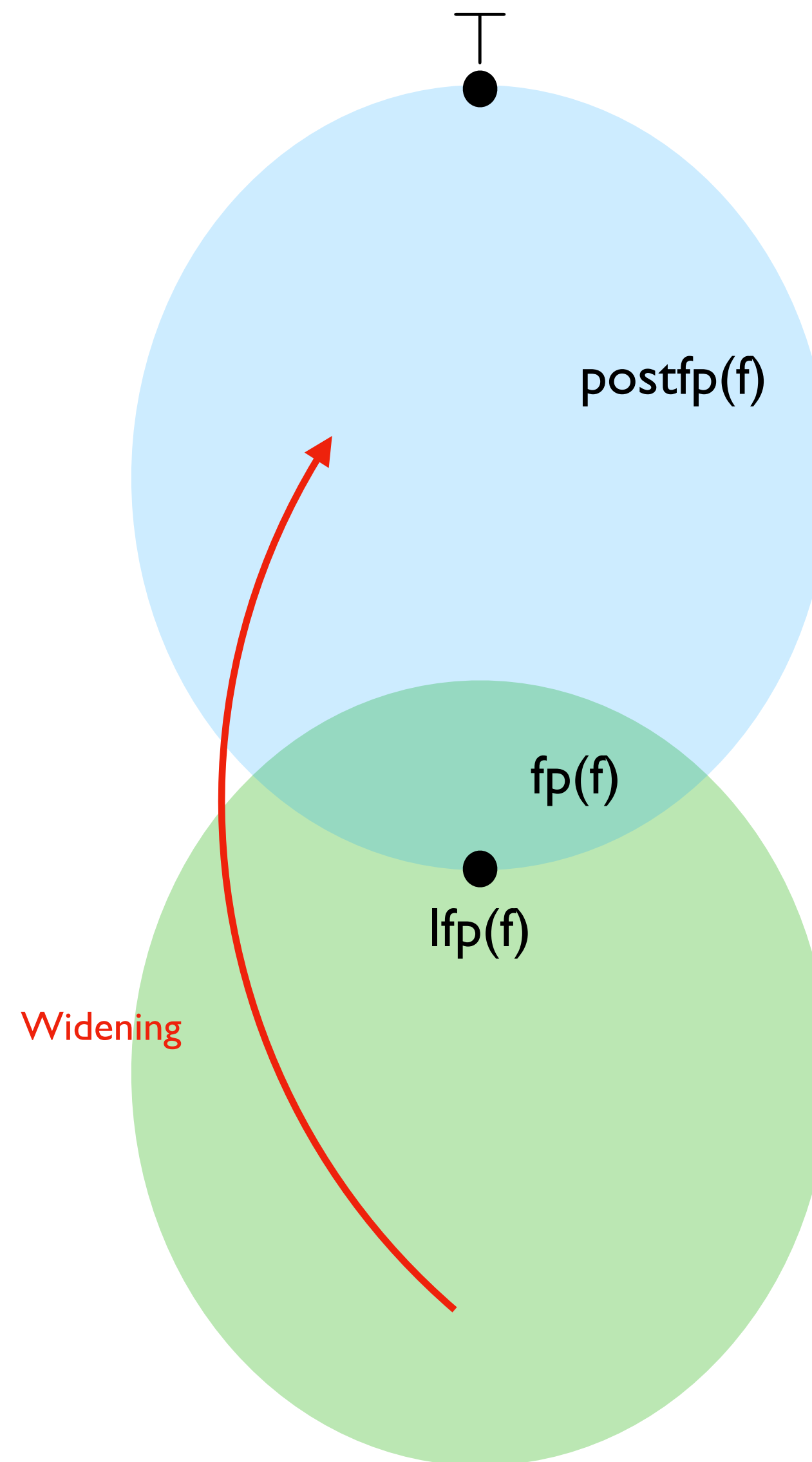
Widening



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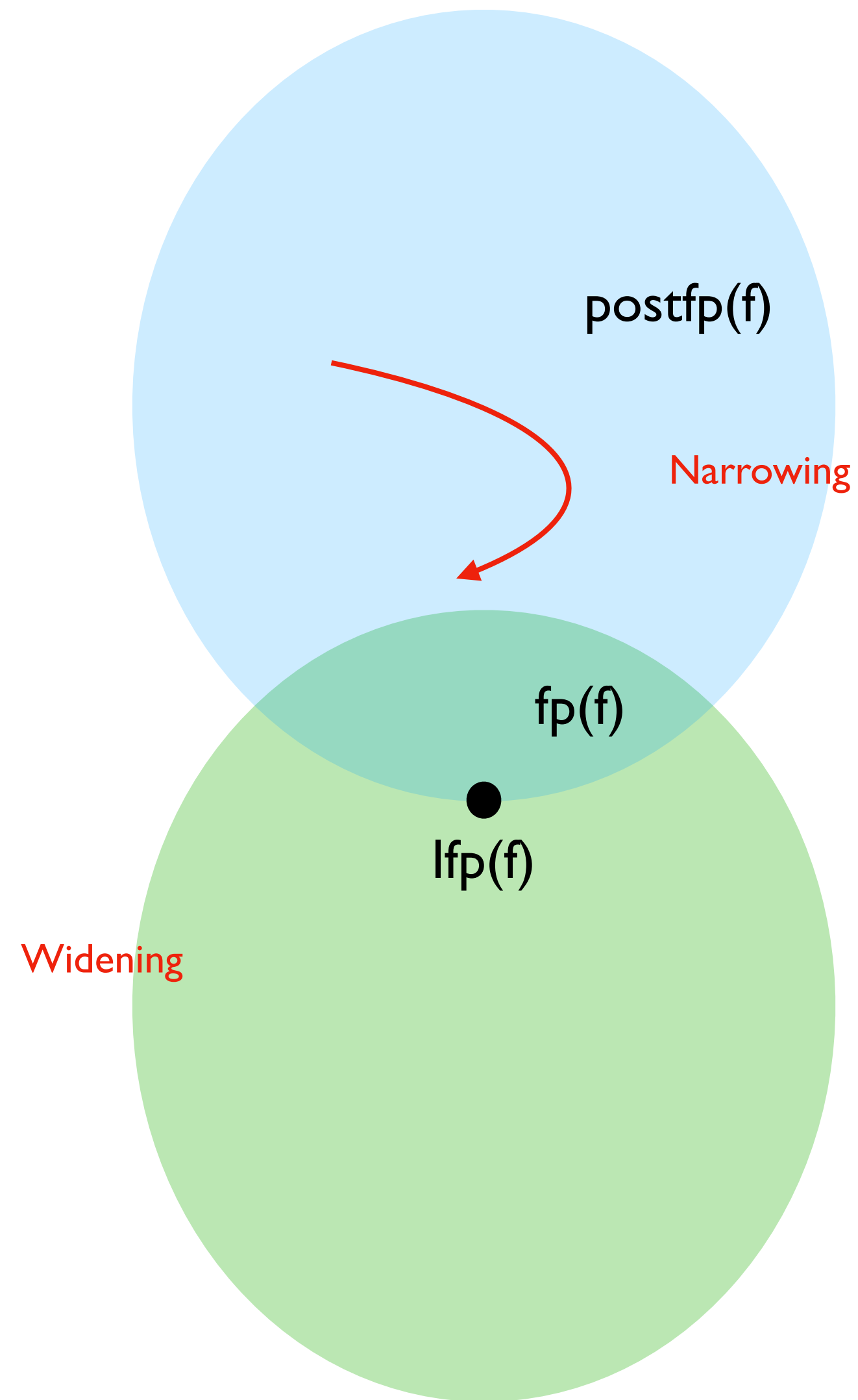


Widening

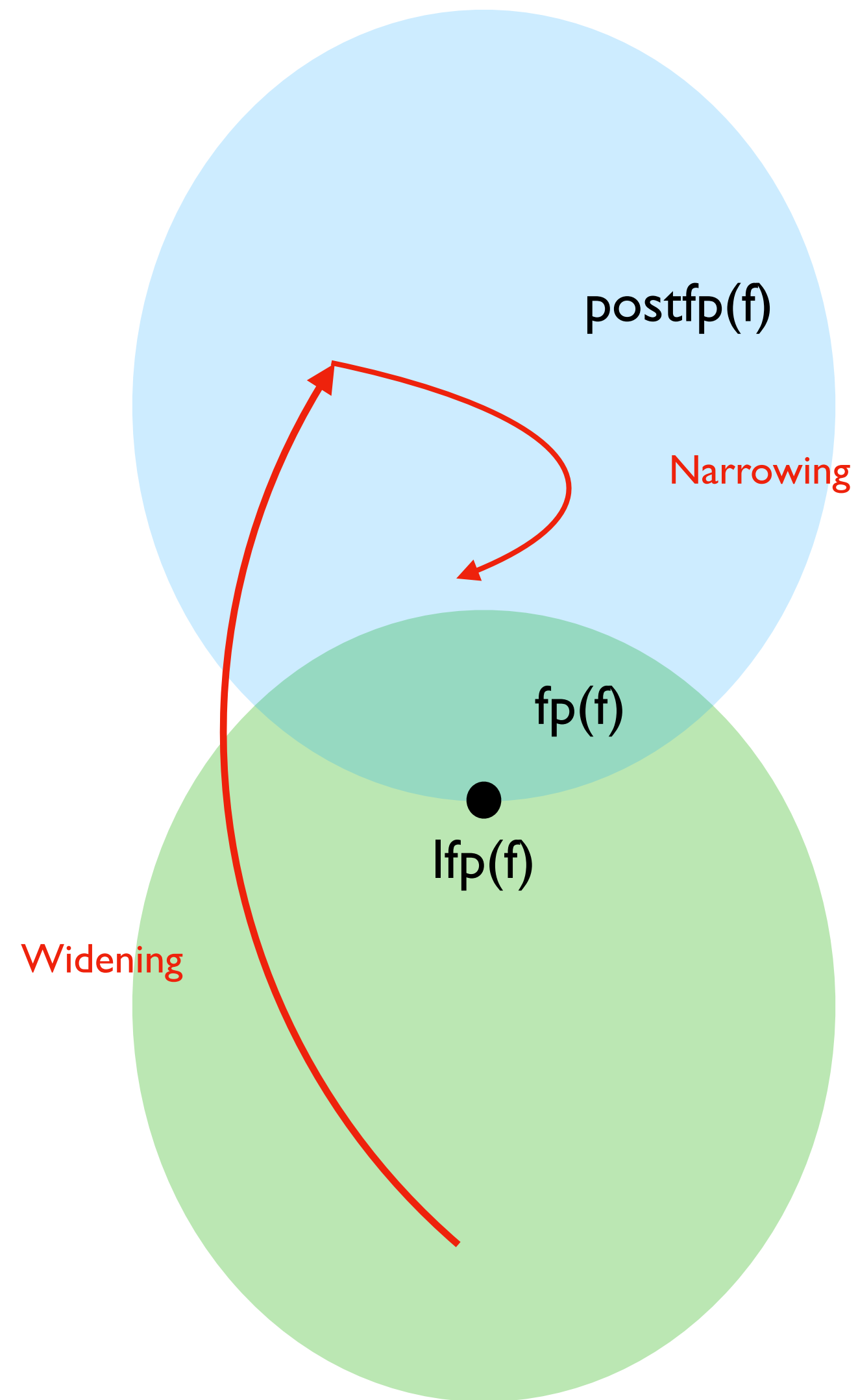


Widening: over-approximates
the $lfp(f)$
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Narrowing



Narrowing



Narrowing

- A binary operator $\triangle : A \times A \rightarrow A$ is a narrowing operator in an abstract domain $\langle A, \sqsubseteq \rangle$ if
 - $\forall x, y \in A : x \sqcap y \sqsubseteq x \triangle y \sqsubseteq x$
 - if ensures convergence: for any sequence $(y^i)_{i \in \mathbb{N}}$ in A , the sequence $(x^i)_{i \in \mathbb{N}}$ computed as $x^0 = y^0, x^{i+1} = x^i \triangle y^{i+1}$ stabilizes in finite time, namely
 - $\exists k \geq 0. x^{k+1} = x^k$

Narrowing for intervals

$$[a, b] \triangle [c, d] = \left[\begin{cases} c & \text{if } a = -\infty \\ a & \text{otherwise} \end{cases}, \begin{cases} d & \text{if } b = +\infty \\ b & \text{otherwise} \end{cases} \right]$$

Narrowing

Where to apply it?

- At the widening points
- starting from the post-fixpoint computed applying widening

$$F^\# = \begin{cases} x_i = x_i \triangle f_i^\#(x_1, \dots, x_m) & \text{if } i \in WP \\ x_i = x_i \sqcap f_i^\#(x_1, \dots, x_m) & \text{otherwise} \end{cases}$$

Ascending and descending phases

